STA 336: Statistical Machine Learning

NFL Wide Receiver Contract Projection

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Abstract

This project studies whether public data can support useful projections for the 2026 veteran wide receiver contract market. The central questions are whether contract term can be predicted, whether annual average value (AAV) can be predicted as a percentage of the NFL salary cap, and which player, contract, production, and market factors appear most important in those predictions. The analysis uses Spotrac wide receiver contract exports as the contract spine and enriches those records with nflverse player, roster, combine, production, snap, Next Gen Stats, team, and schedule data, plus ESPN transaction dates when a likely signing match is available (Spotrac 2026; nflverse 2026; ESPN 2026). The modeling design separates contract term from annual price because the two outcomes represent different market decisions. Term measures commitment, while cap-share AAV measures price conditional on that commitment. The strongest predictive pair uses XGBoost for both tasks (Chen and Guestrin 2016), while lasso multinomial logistic regression and ridge regression provide interpretable audit models (Friedman, Hastie, and Tibshirani 2010). On the signed 2026 validation set, the selected term model achieved log loss of 0.537, Brier score of 0.057, and accuracy of 82.4%; the selected compensation model achieved RMSE of 0.0057, MAE of 0.0026, and R^2 of 0.903. Applied to 163 unsigned 2026 wide receivers, the model projects a short-term market where every scored player has a modal one-year contract, the mean one-year probability is 85.8%, and the median expected annual value is $1.11M. Player-level scenario outputs are available in the companion dashboard.

Introduction and Background

NFL wide receiver contracts are difficult to compare because the headline value of a deal hides several distinct market decisions. A one-year prove-it contract, a two-year role-player contract, and a four-year star extension can all belong to the same position group, yet they imply different levels of team commitment and player leverage. For readers less familiar with football, a wide receiver is an offensive player whose primary role is to catch passes. Contract term is the reported number of contract years, and average annual value, or AAV, is total reported value divided by term. Because the NFL salary cap changes substantially over time, this project models AAV as a share of the league cap at signing rather than as raw dollars. An annual salary of $10 million carries different meaning under a league cap of $120 million than under a league cap of $300 million, so cap share gives a cleaner cross-era target. Rookie and rookie-adjacent deals are excluded because they are governed by a different pricing mechanism; keeping them in the same sample would mix the veteran free-agent market with the rookie wage scale.

The project is organized around three research questions. Can the term of a veteran wide receiver contract be predicted from public information available before signing? Can the contract's AAV as a percentage of the salary cap be predicted with useful accuracy? Which contract-history, career-stage, production, team-context, and market variables explain those predictions? These questions are significant because public contract projections are usually presented as single dollar values, while actual negotiations are closer to menus of possible terms and prices. This motivates the two-model architecture used throughout the project. The term model estimates probabilities over one, two, three, four, and five-or-more-year outcomes, and the compensation model estimates cap-share AAV under each supplied term scenario. The final output is therefore a probability-weighted set of contract paths rather than only a ranked list of dollar forecasts.

Data and Methods

The contract outcome data come from Spotrac wide receiver contract exports grouped by decade (Spotrac 2026). Those exports provide player names, teams, signing ages, start and end years, reported term, total value, average salary, cap share at signing, and guarantee-related fields. The contract records are enriched with nflverse player references, rosters, combine measurements, regular receiving statistics, snap counts, Next Gen receiving metrics, team offensive context, and schedules through `nflreadr` (nflverse 2026). ESPN transaction data are used to recover signing dates when possible (ESPN 2026) so each contract uses only information available before signing; unmatched dates use a March 15 offseason proxy that is recorded in the data. After cleaning, the modeling universe contains 2,796 signed veteran contracts from 2002-2025, 91 signed 2026 validation contracts, and 163 unsigned 2026 free-agent candidates.

Cleaning is designed so that each row represents one veteran contract and each predictor describes the player before signing. The workflow standardizes names, parses contract economics, converts AAV percent to decimal cap share, repairs implausible end years, removes yearly rows already covered by an earlier multi-year deal, and builds prior-contract features. Player matching uses normalized names, age, rookie-season plausibility, roster history, and position context. Recent performance is based on the two most recent fully completed seasons before signing, with the most recent season weighted twice as heavily and missingness tracked with availability indicators rather than treated as a real zero.

The modeling recipe is implemented with tidymodels (tidymodels 2026). Identifiers, audit fields, signing team, and current-contract information are excluded because they would either leak the answer or fail to generalize to unsigned players. Preprocessing handles unknown and new categorical levels, imputes missing predictors, removes zero-variance terms, and normalizes numeric inputs. The term model predicts one, two, three, four, or five-plus years, with longer contracts grouped together because they are rare. The compensation model predicts cap-share AAV under a supplied term scenario. Model selection begins with an untuned baseline screen. The predictive side covers random forest, XGBoost, LightGBM, and neural networks. The interpretable side covers multinomial logistic regression for term, ordinary linear regression for AAV%, ridge, lasso, elastic net, decision trees, and null models. Table 1 summarizes the best baseline result in each task-by-model-role lane. We then fine-tune XGBoost for both predictive tasks, lasso multinomial regression for term, and ridge regression for AAV%. Model selection uses 2002-2025 contracts for training and signed 2026 contracts for validation, with five grouped folds by player used during tuning so the same player's contract history does not leak across folds.

Table 1: Best untuned baseline results by task and model role.

Task	Model Role	Best Baseline	Validation Evidence
Term	Predictive	XGBoost	Log loss 0.560; Brier 0.058; acc. 81.3%
Term	Interpretable	Lasso	Log loss 0.588; Brier 0.060; acc. 81.3%
AAV%	Predictive	XGBoost	RMSE 0.0061; MAE 0.0032; R-squared 0.882
AAV%	Interpretable	Ridge	RMSE 0.0064; MAE 0.0040; R-squared 0.871

Results and Discussion

Figure 1 shows the main forecast pattern. The term task has to be evaluated carefully because the signed 2026 validation set is heavily concentrated in one-year contracts. A model can look accurate by predicting one year for nearly everyone, but that does not mean its probabilities are calibrated or useful for comparing players. For that reason, the selected term model is judged primarily with log loss and Brier score, which reward better probability estimates. After tuning, the selected XGBoost term model improves to log loss 0.537 and Brier 0.057, while the selected XGBoost compensation model explains most held-out variation in cap-share AAV. Its RMSE of 0.0057 represents about 0.57 percentage points of the cap, or roughly $1.7 million at the implied 2026 cap. The first research question therefore receives a qualified answer: contract term can be predicted, but a probability distribution over terms is more useful than a single predicted class. The second research question has a clearer answer: cap-share AAV is meaningfully predictable once the model conditions on term.

The unsigned 2026 wide receiver group is projected to be overwhelmingly short-term. All 163 scored players have one year as their modal term, and the average one-year probability is 85.8%. Price is less compressed than term. The mean expected AAV is $1.49M, the median is $1.11M, and the maximum is $11.52M, so the projected market is right-skewed. Deebo Samuel, Jauan Jennings, and Keenan Allen rise to the top because they combine stronger recent or prior production signals with nontrivial probabilities of multi-year outcomes. Even for those players, however, the modal forecast remains one year. Their expected values should be read as probability-weighted scenario summaries rather than direct predictions of long contracts. Table 2 lists the highest expected-AAV cases and shows how much probability remains on two-or-more-year outcomes.

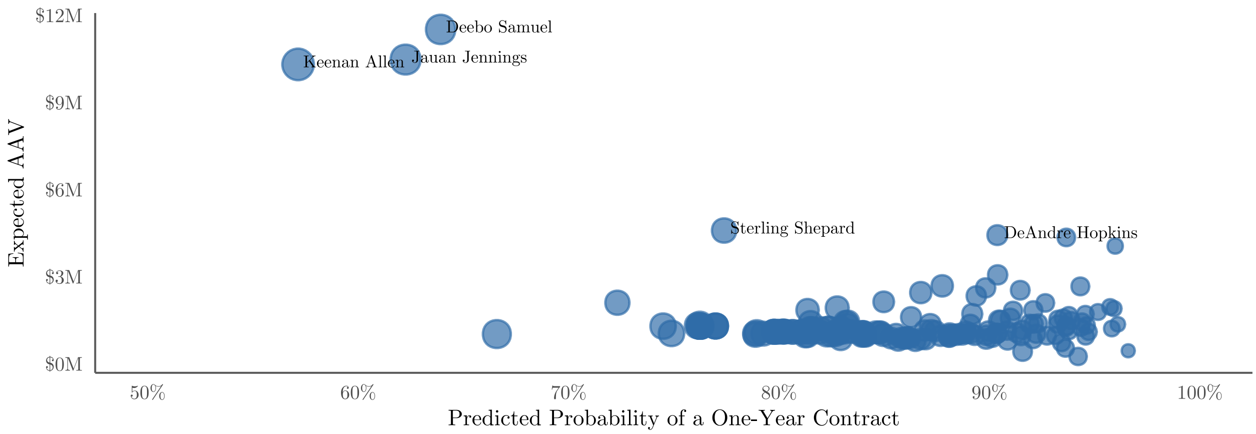

Figure 1: One-year probability versus expected AAV for unsigned 2026 wide receivers.

Table 2: Highest expected AAV projections from the predictive model pair.

Player	Expected Term	P(1 Year)	P(2+ Years)	Expected AAV
Deebo Samuel	1.66	63.9%	36.1%	$11.52M
Jauan Jennings	1.71	62.3%	37.7%	$10.48M
Keenan Allen	1.85	57.1%	42.9%	$10.31M
Sterling Shepard	1.34	77.4%	22.6%	$ 4.59M
DeAndre Hopkins	1.15	90.4%	9.6%	$ 4.43M

The third research question concerns what information is built into the forecasts. The predictive and interpretable models point to the same broad distinction. Term is driven most strongly by contract history and career stage. The boosted term model leans on observed contract number, years of experience, player-match quality, previous contract timing, age, and previous market value. In the lasso audit model, one-year scores are shaped by age, previous-contract timing, and recent game availability; longer-term classes retain production, team-context, and continuity signals. That means the model treats term as a market-commitment decision first and a pure receiving-production decision second. Compensation is tied more directly to receiving value. Weighted receiving yards, receiving yards per game, first downs per game, target share, prior cap share, and the supplied term scenario dominate the boosted importance table. In the ridge audit model, the largest coefficients are term indicators and previous-contract market variables, followed by receiving and snap-context features. The two outcomes are related but not interchangeable; term and price are predictable from different kinds of information.

Player-level term-price menus are available in the companion dashboard.

Limitations and Future Work

Limitations mostly come from measurement and private context. Proxy signing dates can blur the pre-signing window, and name/roster matching is weakest for fringe players. Sparse pre-2020 route data make target volume hard to separate from route skill; route participation, separation, target depth/quality, and injury availability would sharpen that split. Public contracts omit guarantees, incentives, option years, medical risk, cap room, depth-chart need, and agent leverage, so AAV is incomplete. Future work should automate Spotrac refreshes, model guarantees/incentives, add interval and calibration checks, and score cap-and-need team scenarios.

Appendix

Appendix A: Data Construction

The full cleaning workflow begins with Spotrac contract exports, keeps wide receiver rows, standardizes names and teams, parses contract economics, converts cap share to a decimal, buckets five-plus-year contracts, removes entry-like deals, and removes yearly rows already contained inside a prior multi-year contract. The contract spine is enriched with player-reference, roster, combine, regular-stat, snap-count, Next Gen receiving, team, and schedule data. Transaction dates are matched from ESPN by player, team, and contract clues when possible; otherwise the workflow uses the offseason proxy described in the main paper. Tables 3 and 4 summarize the cleaned data splits and feature coverage.

Table 3: Modeling data after cleaning.

Split	Role	Rows	Players	Start Years
Train	Historical signed contracts used for fitting and tuning	2,796	1,305	2002-2025
Validate	Signed 2026 contracts used for held-out model comparison	91	91	2026
Test	Unsigned 2026 free-agent candidates scored by the final models	163	163	2026

Table 4: Training coverage for major external feature families.

Feature Family	Training Coverage
Observed ESPN signing date	60.1%
NFL combine	45.4%
Regular receiving stats	65.8%
Snap usage	51.3%
Next Gen receiving	15.8%
Team context	68.8%
Market context	100.0%

Figure 2 explains why the paper emphasizes probability quality. One-year contracts are 61.0% of the training data and 83.5% of the validation data.

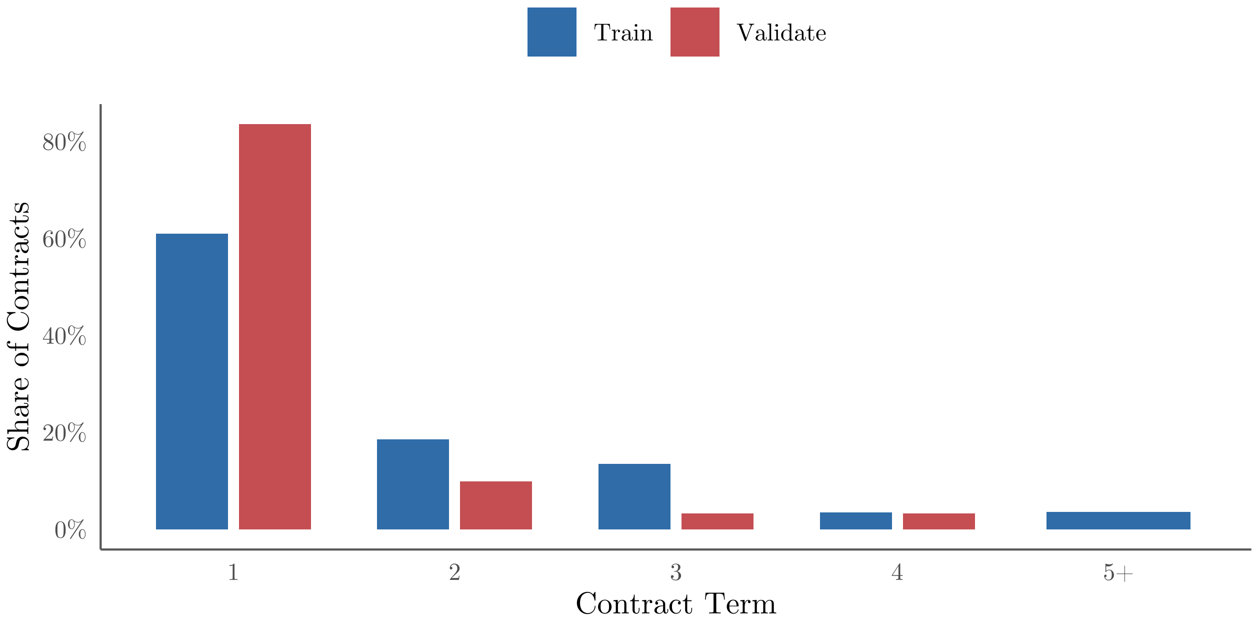

Figure 2: Term distribution in the training and validation data. Five means five or more years.

Appendix B: Model Comparison and Tuning

The baseline stage compared null models, linear and penalized generalized linear models, neural networks, decision trees, random forests, XGBoost, and LightGBM where available. Tuning then focused on the strongest predictive and interpretable candidates from the baseline screen. XGBoost used Bayesian optimization; the penalized models used regularization grids. Table 5 reports the selected tuned comparison used in the paper.

Table 5: Tuned model comparison on the signed 2026 validation set.

Task	Model	Validation Metric 1	Validation Metric 2	Validation Metric 3	Cross-Validation Metric
Term	XGBoost	Log loss = 0.537	Brier = 0.057	Accuracy = 82.4%	CV log loss = 0.788
Term	Lasso multinomial	Log loss = 0.592	Brier = 0.060	Accuracy = 81.3%	CV log loss = 0.832
AAV%	XGBoost	RMSE = 0.0057	MAE = 0.0026	R-squared = 0.903	CV RMSE = 0.0097
AAV%	Ridge linear	RMSE = 0.0061	MAE = 0.0039	R-squared = 0.883	CV RMSE = 0.0105

Tables 6 and 7 give the full successful baseline screens for the term and compensation tasks.

Table 6: Baseline term models ordered by validation log loss.

Model	Log Loss	ROC AUC	Brier	Accuracy
XGBoost	0.560	0.666	0.058	81.3%
Random forest	0.569	0.673	0.059	80.2%
Lasso	0.588	0.645	0.060	81.3%
Elastic net	0.613	0.618	0.061	81.3%
LightGBM	0.652	0.622	0.062	82.4%
Ridge	0.710	0.601	0.065	81.3%
Null majority	0.757	0.500	0.072	83.5%
Multinomial logistic	1.004	0.520	0.075	79.1%
Neural net	1.170	0.546	0.116	72.5%
Decision tree	1.374	0.631	0.069	79.1%

Table 7: Baseline compensation models ordered by validation RMSE.

Model	RMSE	MAE	R-squared
XGBoost	0.0061	0.0032	0.882
Random forest	0.0062	0.0032	0.876
Ridge	0.0064	0.0040	0.871
LightGBM	0.0066	0.0035	0.862
Linear regression	0.0080	0.0048	0.798
Decision tree	0.0104	0.0041	0.710
Elastic net	0.0108	0.0064	0.741
Neural net	0.0110	0.0070	0.616
Lasso	0.0138	0.0083	0.669
Null mean	0.0177	0.0118	NA

Table 8 records the selected XGBoost settings used for the final predictive models.

Table 8: Selected XGBoost hyperparameters.

Model	mtry	Trees	Minimum N	Tree Depth	Learning Rate	Loss Reduction	Sample Size
Term XGBoost	92	1251	10	4	0.005249	0.00016	0.945583
Compensation XGBoost	57	473	2	3	0.078861	0.00001	0.772218

Appendix C: Interpretation Artifacts

The XGBoost models are interpreted through global gain importance. Importance is not a causal effect or a signed marginal coefficient; it reports which transformed predictors most improved split quality across the fitted trees. Figures 3 and 4 show the predictive models' highest-gain features.

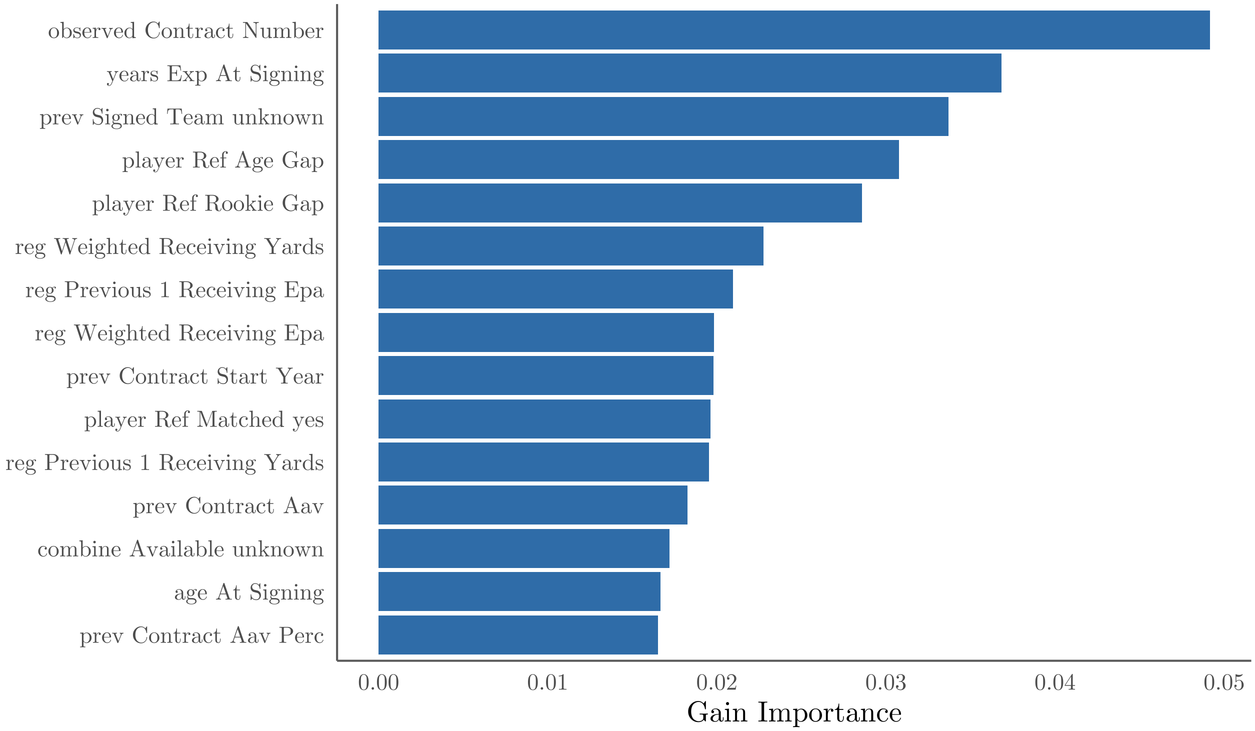

Figure 3: Top gain-importance features for the predictive term model.

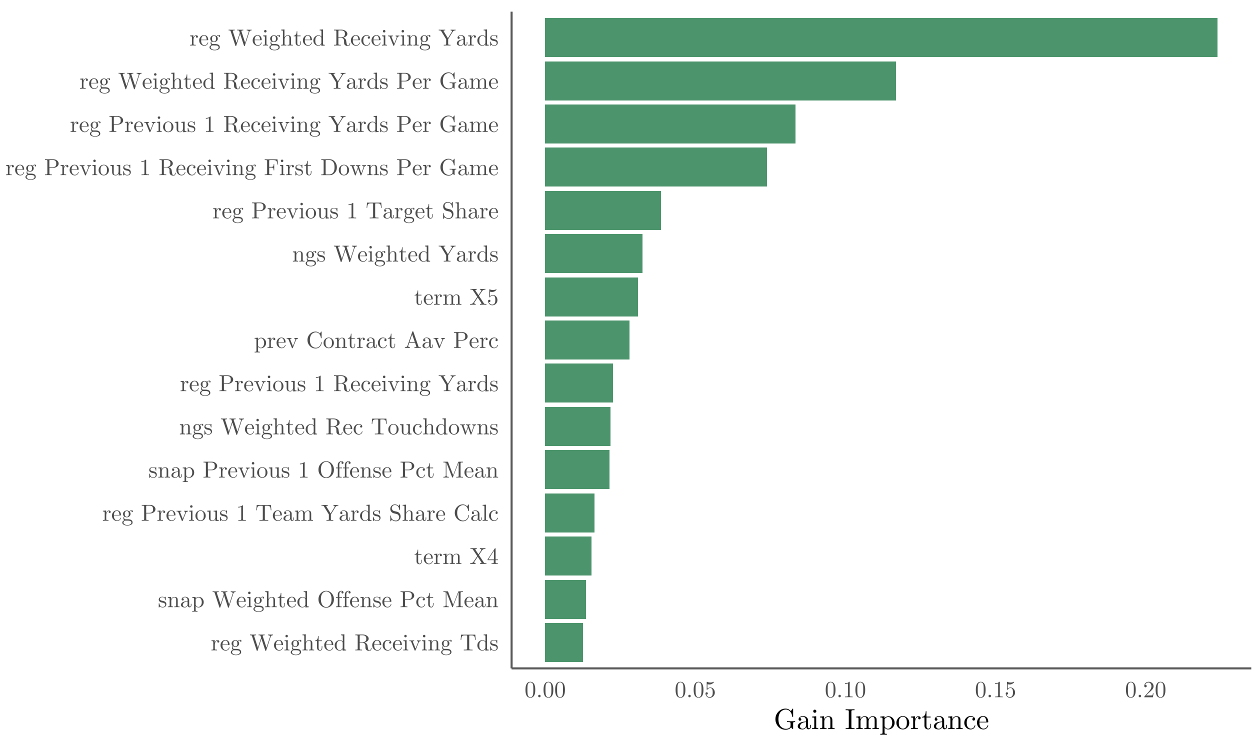

Figure 4: Top gain-importance features for the predictive conditional AAV% model.

Tables 9 and 10 summarize the lasso audit model. The sparsity table shows how many predictors remain active by term class, while the coefficient table shows the strongest retained signals within each class.

Table 9: Class-specific sparsity in the lasso multinomial term model.

Term Class	Non-Zero Coefficients
1	67
2	45
3	22
4	9
5+	13

Table 10: Five largest absolute nonzero lasso coefficients by term class.

Term Class	Feature	Coefficient	Absolute Value
1	age At Signing	0.319	0.319
1	prev Contract Start Year	0.281	0.281
1	reg Previous 1 Games	-0.189	0.189
1	prev Contract End Year	0.167	0.167
1	reg Previous 2 Games	-0.166	0.166
2	prior Team Same unknown	0.253	0.253
2	gap Since Prev Contract End	-0.177	0.177
2	reg Weighted Games	-0.162	0.162
2	undrafted Flag yes	0.137	0.137
2	reg Previous 1 Receiving Yards Per Game	-0.122	0.122
3	prev Signed Team unknown	0.662	0.662
3	player Ref Age Gap	0.372	0.372
3	combine Available unknown	0.269	0.269
3	team Weighted Receiving Air Yards	0.161	0.161
3	ngs Previous 1 Avg Yac Above Expectation	0.110	0.110
4	prior Team Same yes	0.183	0.183
4	reg Previous 1 Receiving Epa	0.169	0.169
4	snap Weighted Offense Snaps	0.104	0.104
4	ngs Weighted Yards	0.102	0.102
4	team Previous 1 Receiving Air Yards	-0.078	0.078
5+	start Year	-0.403	0.403
5+	reg Weighted Team Yards Share Calc	0.221	0.221
5+	reg Previous 1 Receiving First Downs Per Game	0.196	0.196
5+	reg Previous 2 Receiving Epa Per Game	0.140	0.140
5+	market Previous 2 Contract Count	-0.085	0.085

Table 11 gives the largest absolute coefficients from the ridge compensation audit model.

Table 11: Largest absolute coefficients in the ridge compensation model.

Feature	Coefficient	Absolute Value
term X5	0.004123	0.004123
term X4	0.002808	0.002808
term X3	0.001850	0.001850
prev Signed Team unknown	-0.001230	0.001230
prev Contract Aav	0.001094	0.001094
prev Contract Aav Perc	0.001079	0.001079
ngs Previous 2 Yards	0.001076	0.001076
ngs Delta Avg Separation	-0.000977	0.000977
ngs Previous 1 Yards	0.000937	0.000937
snap Previous 2 Offense Pct Mean	0.000926	0.000926
prior Team Same yes	0.000908	0.000908
reg Years Available	-0.000882	0.000882
reg Previous 2 Available yes	-0.000868	0.000868
reg Weighted Games	-0.000859	0.000859
reg Previous 1 Receiving Yards	0.000828	0.000828

Appendix D: Additional Forecast Outputs

Table 12 expands the main-paper projection table to the ten highest expected-AAV players. Figure 5 shows how predicted AAV changes when the same player is evaluated under different term scenarios, with point size representing the term probability assigned by the term model.

Table 12: Ten highest expected AAV projections from the predictive model pair.

Player	Age	Previous Team	Expected Term	P(1 Year)	P(2+ Years)	Expected AAV	Modal AAV
Deebo Samuel	30	WAS	1.66	63.9%	36.1%	$11.52M	$10.16M
Jauan Jennings	28	SF	1.71	62.3%	37.7%	$10.48M	$ 9.05M
Keenan Allen	33	LAC	1.85	57.1%	42.9%	$10.31M	$ 8.73M
Sterling Shepard	33	TB	1.34	77.4%	22.6%	$ 4.59M	$ 4.19M
DeAndre Hopkins	33	BAL	1.15	90.4%	9.6%	$ 4.43M	$ 4.19M
Adam Thielen	35	CAR	1.09	93.7%	6.3%	$ 4.35M	$ 4.22M
Josh Reynolds	31	NYJ	1.05	96.0%	4.0%	$ 4.06M	$ 4.01M
Ray-Ray McCloud	29	ATL	1.13	90.4%	9.6%	$ 3.07M	$ 2.90M
Gabriel Davis	27	BUF	1.20	87.8%	12.2%	$ 2.69M	$ 2.36M
Elijah Moore	25	BUF	1.09	94.3%	5.7%	$ 2.67M	$ 2.51M

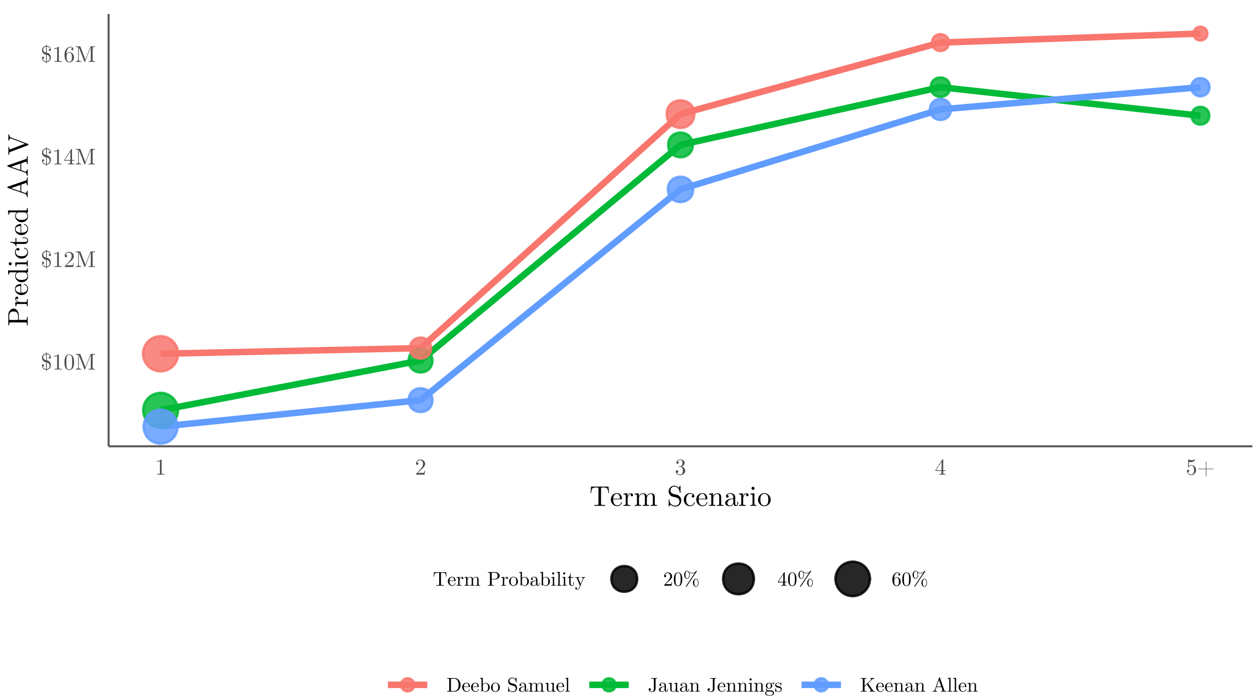

Figure 5: Scenario-level AAV paths for the three highest expected-AAV players.

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